

CUET PG CALCULUS

COMPLETE ANALYSIS



Function [04]

01. Let $f : [2, \infty] \rightarrow \mathbb{R}$ be the function defined by $f(x) = x^2 - 4x + 5$, then the range of f [CUET 2022]
 (1) $\mathbb{R}$ (2) $[1, \infty)$ (3) $[4, \infty)$ (4) $(-\infty, 0]$

02. Let $A = \{1, 2, 3\}$ and consider the relation $R = \{(1, 1), (2, 2), (3, 3), (1, 2), (1, 3)\}$ then R is [CUET 2022]
 (1) Reflexive but not symmetric (2) Reflexive but not transitive
 (3) Symmetric and transitive (4) Equivalence relation

03. Let b be a positive integer and $R = \{(a, b) \in \mathbb{Z} \times \mathbb{Z} \mid a - b = nm \text{ for some } m \neq 0 \in \mathbb{Z}\}$ [CUET 2022]
 (1) Reflexive on $\mathbb{Z}$ (2) Symmetric (3) Transitive (4) Equivalence relation of $\mathbb{Z}$

04. Match List – I with List – II [CUET 2024]
List – I (Function) **List – II (Range)**

- | | |
|---|--|
| (A) $y = \frac{1}{2 - \sin 3x}$ | (I) $\left(1, \frac{7}{3}\right]$ |
| (B) $y = \frac{x^2 + x + 2}{x^2 + x + 1}, x \in \mathbb{R}$ | (II) $\left[\frac{\pi}{2}, \pi\right) \cup \left(\pi, \frac{3\pi}{2}\right]$ |
| (C) $y = \sin x - \cos x$ | (III) $\left[\frac{1}{3}, 1\right]$ |
| (D) $y = \cot^{-1}(-x) - \tan^{-1} x + \sec^{-1} x$ | (IV) $[-\sqrt{2}, \sqrt{2}]$ |

Choose the correct answer from the options given below:

- | | |
|--|--|
| (1) [(A – III); (B – I); (C – IV); (D – II)] | (2) [(A – III); (B – II); (C – IV); (D – I)] |
| (3) [(A – II); (B – III); (C – I); (D – IV)] | (4) [(A – II); (B – III); (C – IV); (D – I)] |



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Limits [05]

01. Match List I with List II

[CUET 2023]

LIST I

LIST II

A. Value of $\lim_{x \rightarrow 0} \left(\frac{\sin x}{x} \right)^{\frac{\sin x}{x - \sin x}}$ is

I. e^3

B. Value of $\lim_{x \rightarrow 0} \int_0^x \frac{\sin t^2 dt}{x^2}$ is

II. 0

C. Value of $\lim_{x \rightarrow 0} (e^{2x} + x)^{\frac{1}{x}}$ is

III. 1

D. Value $\lim_{x \rightarrow a} \frac{\log(x - a)}{\ln(e^x - e^a)}$ of

IV. e^{-1}

Choose the correct answer from the options given below:

(1) A-II, B-III, C-I, D-IV

(2) A-II, B-IV, C-III, D-I

(3) A-IV, B-II, C-III, D-I

(4) A-IV, B-II, C-I, D-III

02. $\lim_{x \rightarrow 0} \frac{\sqrt{1 - \cos 2x}}{x} =$

[CUET 2024]

(1) 0

(2) 1

(3) $\sqrt{2}$

(4) $1/2$

03. Match List – I with List – II

[CUET 2024]

List – I

List – II

Function $f(x)$

$f(0)$

(A) $f(x) = \frac{\log(1 + 4x)}{x}$

(I) $\frac{1}{4}$

(B) $f(x) = \frac{\log(4 + x) - \log 4}{x}$

(II) 1

(C) $f(x) = \frac{x}{\sin x}$

(III) 4

(D) $f(x) = \frac{1 - \cos^3 x}{x \sin 2x}$

(IV) $\frac{3}{4}$

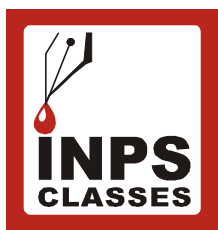
Choose the correct answer from the options given below:

(1) [(A – I); (B – III); (C – IV); (D – II)]

(2) [(A – I); (B – III); (C – II); (D – IV)]

(3) [(A – III); (B – I); (C – II); (D – IV)]

(4) [(A – III); (B – I); (C – IV); (D – II)]



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04. Math List - I with List - II

[CUET 2025]

List – I

List – II

(A) $\lim_{x \rightarrow 0} (1 + 2x)^{\frac{1}{x}}$

(i) e^6

(B) $\lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right)^x$

(ii) e^2

(C) $\lim_{x \rightarrow 0} (1 + 5x)^{\frac{1}{x}}$

(iii) e

(D) $\lim_{x \rightarrow \infty} \left(1 + \frac{3}{x}\right)^{2x}$

(iv) e^5

Choose the correct answer from the options given below:

(1) (A) – (iv), (B) – (i), (C) – (iii), (D) – (ii) (2) (A) – (i), (B) – (iv), (C) – (iii), (D) – (ii)

(3) (A) – (ii), (B) – (iii), (C) – (iv), (D) – (i) (4) (A) – (iii), (B) – (ii), (C) – (i), (D) – (iv)

05. The value of $\lim_{x \rightarrow \infty} \left(1 + \frac{2}{3x}\right)^x$ is:

[CUET 2025]

(1) e

(2) e^2

(3) $e^{2/3}$

(4) $e^{1/3}$



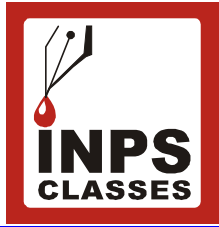
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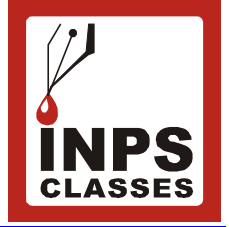
Continuity & Differentiability [08]

01. Let $f(x) = ||x| - 1|$, then point(s) where $f(x)$ is not differentiable is (are): [CUET 2022]
 (1) $0, \pm 1$ (2) ± 1 (3) 0 (4) 1
02. The function $f(x) = \frac{[\ln(1 + \alpha x) - \ln(1 - \beta x)]}{x}$ is not defined at $x = 0$. What value may be assigned to f at $x = 0$, so that it is continuous? [CUET 2022]
 (1) $a + b$ (2) $a - b$ (3) $b - a$ (4) $\ln a + \ln b$
03. Given below are two statements : One is labelled as Assertion A and the other is labelled as Reason R.
Assertion A : $f(x) = \tan^2 x$ is continuous at $x = \pi/2$
Reason R : $g(x) = x^2$ is continuous at $x = \pi/2$ [CUET 2023]
 In the light of the above statements, choose the correct answer from the options given below :
 (1) Both A and R are true and R is the correct explanation of A
 (2) Both A and R are true but R is not the correct explanation of A
 (3) A is true but R is false
 (4) A is false but R is true
04. The point(s) at which function f is given by $f(x) = \begin{cases} \frac{x}{|x|}; & x < 0 \\ -1; & x \geq 0 \end{cases}$ is continuous is/are [CUET 2023]
 (1) $x \in \mathbb{R}$ (2) $x = 0$ (3) $x \in \mathbb{R} \setminus \{0\}$ (4) -1 and 1
05. Which of the following functions is differentiable at $x = 0$? [CUET 2023]
 (1) $\cos(|x|) + |x|$ (2) $\cos(|x|) - |x|$ (3) $\sin(|x|) + |x|$ (4) $\sin(|x|) - |x|$
06. The function $f(x) = [x]^n$, integer $n \geq 2$ (where $[y]$ is the greatest integer less than or equal to y), is discontinuous at all point of [CUET 2024]
 (1) real number (2) all non-integer real numbers (3) only at zero (4) integers
07. A function $f(x)$ is defined as $f(x) = \begin{cases} \frac{1 - \cos 4x}{x^2} & ; x < 0 \\ a & ; x = 0 \\ \frac{\sqrt{x}}{\sqrt{(16 + \sqrt{x})} - 4} & ; x > 0 \end{cases}$
 if the function $f(x)$ is continuous at $x = 0$, then the value of a is: [CUET 2024]
 (1) 4 (2) 6 (3) 8 (4) 10



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08. If $f(x) = \begin{cases} x \sin\left(\frac{1}{x}\right), & x \neq 0 \\ 0, & x = 0 \end{cases}$ then $f(x)$ is

[CUET 2025]

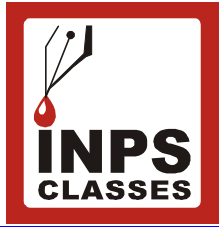
(1) continuous for all $x \in \mathbb{R}$

(2) continuous at 0, 1 only

(3) not continuous at 1

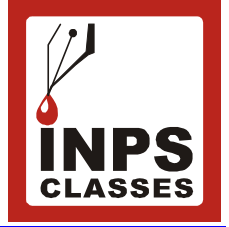
(4) not continuous at 0

INPS



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Differentiation [00]

INPS



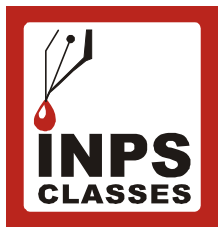
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Application of Derivatives [06]

01. A spring is being moved up and down. An object is attached to the end of the spring that undergoes a vertical displacement. The displacement is given by the equation $y = 3.50 \sin t + 1.20 \sin 2t$. Find the first two values of t (in seconds) for which $y = 0$. [CUET 2022]
- (1) $t = 0, \frac{\pi}{4}$ (2) $t = 0, \frac{\pi}{2}$ (3) $t = 0, \pi$ (4) $t = 0, \frac{\pi}{6}$
02. A ball is thrown off the edge of a building at an angle of 60° and with the initial velocity of 5 meters per second. The equation that represents the horizontal distance of the ball x is $x = v_0 (\cos \theta) t$, where v_0 is the initial velocity, θ is the angle at which it is thrown and t is the time in seconds. About how far will the ball travel in 10 seconds? [CUET 2022]
- (1) $25\sqrt{3}\text{m}$ (2) $50\sqrt{2}\text{m}$ (3) 25 m (4) $\frac{25}{\sqrt{3}}\text{m}$
03. If f and g are differentiable functions in $(0, 1)$ satisfying $f(0) = 2 = g(1)$, $g(0) = 0$ and $f(1) = 6$, then for some $c \in]0, 1[$. [CUET 2023]
- (1) $2f'(c) = g'(c)$ (2) $2f'(c) = 3g'(c)$ (3) $f'(c) = g'(c)$ (4) $f'(c) = 2g'(c)$
04. If the curve $ay + x^2 = 7$ and $x^3 = y$ cut orthogonally at $(1, 1)$ then the value of a is [CUET 2023]
- (1) 1 (2) 6 (3) -6 (4) 0
05. If $f: \mathbb{R} \rightarrow \mathbb{R}$ defined as $f(x) = x^2 + 1$ then minimum value of $f(x)$ is [CUET 2023]
- (1) 4 (2) 3 (3) 2 (4) 1
06. If the parametric equation of a curve is given by $x = e^t \cos t$ and $y = e^t \sin t$ then the tangent to the curve at the point $t = \frac{\pi}{4}$ makes the angle with the axis of x is [CUET 2024]
- (1) 0 (2) $\frac{\pi}{4}$ (3) $\frac{\pi}{3}$ (4) $\frac{\pi}{2}$



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Indefinite Integration [02]

01. $\int \frac{(x^5 - x)^{\frac{1}{5}}}{x^6} dx$ is equal to (where C is an arbitrary constant)

[CUET 2024]

(1) $\left(1 - \frac{1}{x^4}\right)^{\frac{4}{5}} + C$ (2) $\left(x^4 - \frac{1}{x^4}\right)^{\frac{6}{5}} + C$ (3) $\frac{5}{24} \left(1 - \frac{1}{x^4}\right)^{\frac{6}{5}} + C$ (4) $\frac{5}{24} \left(x^4 - \frac{1}{x^4}\right)^{\frac{6}{5}} + C$

02. $\int \frac{2x+1}{x^2+x+2} dx$ is

[CUET 2025]

(1) $\log(2x+1) + c$, where c is an arbitrary constant

(2) $\log \frac{(2x+1)}{(x^2+x+2)} + c$, where c is an arbitrary constant

(3) $\log(x^2+x+2) + c$, where c is an arbitrary constant

(4) $\log\left(\frac{1}{2}\right) + c$, where c is an arbitrary constant



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Definite Integration [03]

01. Given below are two statements : One is labelled as Assertion A and the other is labelled as Reason R.

Assertion A: $\int_{-3}^3 (x^3 + 5) dx = 30$

[CUET 2023]

Reason R : $f(x) = x^3 + 5$ is an odd function

In the light of the above statements, choose the correct answer from the options given below :

- (1) Both A and R are true and R is the correct explanation of A
 (2) Both A and R are true but R is not the correct explanation of A
 (3) A is true but R is false
 (4) A is false but R is true

02. If $f(a + b - x) = f(x)$ then $\int_a^b xf(x) dx$ is equal to

[CUET 2024]

(1) $\frac{b+a}{2} \int_a^b f(x) dx$ (2) $\frac{b-a}{2} \int_a^b f(x) dx$ (3) $\frac{a+b}{2} \int_a^b f(a+x) dx$ (4) $\frac{a+b}{2} \int_a^b xf(x) dx$

03. Match List – I with List – II

[CUET 2024]

List – I

List – II

(A) $\int_0^{\frac{\pi}{2}} \frac{\sin^4 x}{\sin^4 x + \cos^4 x} dx$

(I) 0

(B) $\int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{1}{1 + \sqrt{\tan x}} dx$

(II) 1

(C) $\int_0^1 x e^x dx$

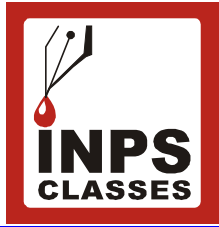
(III) $\frac{\pi}{12}$

(D) $\int_{-1}^1 x^{109} \cos^{88} x dx$

(IV) $\frac{\pi}{4}$

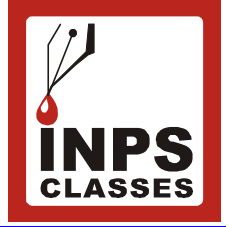
Choose the correct answer from the options given below:

- (1) [(A – IV); (B – III); (C – I); (D – II)] (2) [(A – IV); (B – III); (C – II); (D – I)]
 (3) [(A – III); (B – IV); (C – II); (D – I)] (4) [(A – III); (B – IV); (C – I); (D – II)]



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Area Under Curve [02]

01. The area enclosed between the graphs of $y = x^3$ and the lines $x = 0$, $y = 1$, $y = 8$ is

[CUET 2022]

- (1) 7 (2) 12 (3) $\frac{45}{4}$ (4) $\frac{21}{8}$

02. The area of the region bounded by the curve $y^2 = 4x$ and $x^2 = 4y$ is

[CUET 2024]

- (1) $\frac{16}{3}$ sq.units (2) $\frac{23}{6}$ sq.units (3) $\frac{13}{3}$ sq.units (4) $\frac{28}{5}$ sq.units